Circles, Polygons and Surface — Diagnosing Radius/Diameter and Area/Perimeter Errors

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Two swaps cause most of the wrong answers in this unit. Catch them before you compute.

- **Radius or diameter?** $C = 2\pi r = \pi d$ and $A = \pi r^2$. The area formula *only* takes a radius, so halve a diameter first.
- **Around or inside?** Circumference and perimeter are lengths (m, cm); area is a covering (m^2, cm^2). The unit is the fastest check.
- Use the π key on your calculator and round every decimal answer to the nearest hundredth. Show the formula you chose before you substitute.

1. Warm-up. A circular badge has radius 5 cm. Find the circumference of the badge.

Answer: ________________ cm

2. Halve it first. A circular tabletop is 14 cm across at its widest point — that is its diameter. Find the area of the tabletop.

Answer: ________________ cm^2

3. Find and fix. Rae was asked for the *area* of a circle of radius 6 cm. She wrote $A = 2\pi r = 2\pi(6) \approx 37.70$ and gave 37.70 cm^2 as her answer.

Name the formula Rae actually used and say what quantity it measures. Then find the area she was asked for.

Answer: ________________ cm^2

4. Find and fix. Marco is buying edging to run all the way around a rectangular garden bed that is 9 m long and 4 m wide. He wrote $9 \times 4 = 36$ and ordered 36 m of edging.

Name the measurement Marco actually calculated. Then find how many m of edging the bed really needs.

Answer: ________________ m

5. L-shaped plot. On a coordinate grid, a plot of land has corners at $(0, 0)$, $(10, 0)$, $(10, 4)$, $(4, 4)$, $(4, 7)$ and $(0, 7)$, joined in that order. Each grid unit is one metre.

A landscaper needs the amount of *ground the plot covers*, not the fence around it. Find that amount, in square units.

Answer: ________________ square units

6. Read the given quantity carefully. A round rug measures 1.8 m from one edge, through the centre, to the opposite edge. Trim is sewn around the entire outer edge of the rug. How many m of trim are needed?

Before substituting, write down whether 1.8 is a radius or a diameter.

Answer: ________________ m

7. Work backwards. A circular pool is fenced by exactly 44 m of railing running around its edge. Find the area of the pool's surface.

The number you are given is a length around the circle, so it is not a radius — recover the radius from it first, then use the area formula.

Answer: ________________ m^2

8. Explain. Circle B has exactly twice the radius of circle A.

(a) Explain why circle B's *circumference* is twice circle A's, but its *area* is four times circle A's. Use the formulas in your explanation.
(b) Write a two-part habit a classmate could use on every circle problem: one part about deciding between a length and a covering, one part about checking the unit of the answer.
(c) Explain how that habit would have caught Rae's mistake in problem 3.